

DD_FUL-09 - Equipment Swap Fulfillment

Version: v1.0 **Bundle:** 06_fulfillment **Status:** Draft for review **Primary owner:** Fulfillment / provisioning team **Related DDs:** DD_OSR-RMA-01, DD_OSR-01, DD_OSR-INSTANCE-01, DD_OSR-02, DD_WO-01, DD_WO-01-FLOW-SHIFTING, DD_FUL-03, DD_PROV-INT-01, DD_RLM-CFG-01, DD_BIL-01, DD_EM-CFG-04, FOUNDATION_DROOLS

1. Purpose

FUL-09 defines the fulfillment-side service and provisioning work required when customer equipment is swapped, replaced, upgraded, recovered, or re-bound.

OSR-RMA-01 owns the equipment swap/RMA operational workflow: source/target equipment, contractor route, recovery, RMA state, and stock movement. FUL-09 owns the fulfillment effect: service continuity, network rebinding, provisioning commands, customer subscription impact, HomePass updates, and final activation confirmation.

Use FUL-09 when:

- swap workflow needs old device unbound and new device provisioned;
- HFC modem replacement must preserve or update slot/port authorization;
- GPON ONT replacement must unbind old ONT and bind new ONT at the OLT/HomePass;
- STB/smart card replacement must update entitlement bindings;
- equipment swap is part of relocation/shifting or upgrade;
- swap is completed physically but provisioning/network state needs reconciliation.

Procurement is related only upstream. OSR-02 creates future stock supply. FUL-09 must not call procurement inside a customer swap. Customer jobs consume available/reserved OSR stock.

2. Scope And Boundaries

Capability	Owner
Swap request, recovery logistics, RMA routing, source/target instance state	OSR-RMA-01
SKU quantity reservation/movement	OSR-01
Serialized instance identity/current state/lifecycle	OSR-INSTANCE-01
Upstream stock replenishment/procurement	OSR-02
Work order technician execution	WO-01
Network/service provisioning commands	FUL-09 / FUL-03 / PROV-INT-01
HomePass topology/path update	RLM-CFG-01 via FUL-09/FUL-07 where applicable
Billing charges/deposit delta	BIL-01 / BIL-02
Configurable override approval	EM-CFG-04
Operator-variable swap fulfillment rules	Drools

Boundary rules:

- FUL-09 does not create purchase orders or procure stock.
- FUL-09 does not allocate inventory. It consumes target equipment selected/reserved by OSR-RMA/OSR.
- FUL-09 does not mutate OSR tables directly. It calls OSR-INSTANCE lifecycle APIs and consumes OSR-RMA events.
- FUL-09 does not replace FUL-03; it coordinates equipment-specific provisioning and may call FUL-03/PROV-INT adapters.
- Approval is needed only for risky manual override, forced completion, or exception repair. Normal swap execution follows workflow/rules.

3. Process Overview

```

flowchart LR
    A["OSR-RMA swap ready for fulfillment"] --> B["FUL-09 validates service/equipment context"]
    B --> C["Reserve/provision target path"]
    C --> D["Unbind old equipment"]
    D --> E["Bind/provision new equipment"]
    E --> F["Run reconciliation"]
    F --> G["Notify OSR-RMA fulfillment complete"]
    G --> H["OSR updates lifecycle and closes swap"]

```

For relocation/shifting:

- SUB-WF-RELOCATION creates the shifting operation.
- WO-01 shifting executes field work.
- OSR-RMA/OSR handle equipment recovery/new equipment lifecycle.
- FUL-09 handles the network rebinding and service activation at target.
- OSR-02 procurement is not in the runtime path.

4. Fulfillment Status Model

Status	Meaning
REQUESTED	Swap fulfillment requested by OSR-RMA/WO/SUB flow.
VALIDATED	Service/equipment context validated.
PROVISIONING_OLD_UNBIND	Old device unbind in progress.
PROVISIONING_NEW_BIND	New device bind/provision in progress.
RECONCILING	Network/BSS state being checked.
COMPLETED	Fulfillment complete.
FAILED	Fulfillment failed and can be retried or repaired.
PENDING_APPROVAL	Manual override/force-complete approval open.
CANCELLED	Fulfillment cancelled before network change.

5. Data Model

5.1 equipment_swap_fulfillment

What it is for: stores one fulfillment execution for an equipment swap request.

When to use it: every OSR-RMA swap that requires provisioning/network/service-side changes should create one fulfillment row.

Sample meaning: esf-001 is the fulfillment execution that unbinds old ONT inst-old-001 and binds new ONT inst-new-001 for subscription SUB-700045.

Column	Type	Purpose	Sample
fulfillment_id	TEXT PK	Fulfillment id.	esf-001
operator_code	TEXT	Tenant/operator.	WIK
swap_request_id	TEXT	OSR-RMA swap request.	swap-001
wo_id	TEXT NULL	Work order if field-driven.	wo-shift-001
subscription_id	TEXT	Subscription id.	SUB-700045
account_id	TEXT NULL	Account id.	ACC-900045
customer_id	TEXT NULL	Customer id.	CUS-100045
swap_type	TEXT	DEFECT_SWAP, EQUIPMENT_UPGRADE, RELOCATION_SWAP, TERMINATION_RECOVERY.	DEFECT_SWAP
technology_family	TEXT	GPON, HFC, IPTV, VOICE, MIXED.	GPON
source_instance_id	TEXT NULL	Old equipment instance.	inst-old-001
target_instance_id	TEXT NULL	New equipment instance.	inst-new-001

Column	Type	Purpose	Sample
source_sku_id	TEXT NULL	Old SKU.	WIK-ONT-HUAWEI-OLD
target_sku_id	TEXT NULL	New SKU.	WIK-ONT-HUAWEI-EG8145V5
homepass_id	TEXT NULL	Target/current HomePass.	HP-001
status	TEXT	Fulfillment status.	PROVISIONING_NEW_BIND
created_at	TIMESTAMP	Creation timestamp.	2026-07-01T08:00:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	null

Recommended indexes:

- unique (operator_code, swap_request_id).
- (subscription_id, status).
- (wo_id).

5.2 equipment_swap_provisioning_step

What it is for: stores each provisioning command/check in the swap fulfillment flow.

When to use it: create one row per old-unbind, new-bind, entitlement update, HomePass update, reconciliation, or rollback command.

Sample meaning: esps-001 records the command sent to unbind old ONT from the OLT.

Column	Type	Purpose	Sample
step_id	TEXT PK	Step id.	esps-001
fulfillment_id	TEXT	Parent fulfillment.	esf-001
step_code	TEXT	Step identifier.	GPON_UNBIND_OLD_ONT
target_system	TEXT	Provisioning target.	OLT_NMS
command_ref	TEXT NULL	PROV-INT command id.	prov-cmd-001
status	TEXT	PENDING, SENT, SUCCESS, FAILED, SKIPPED, COMPENSATED.	SUCCESS
request_json	JSONB	Command request snapshot.	{"oldSerial":"OLD123"}
response_json	JSONB NULL	Response snapshot.	{"result":"OK"}
failure_code	TEXT NULL	Failure code.	null
started_at	TIMESTAMP NULL	Start time.	2026-07-01T08:02:00Z
completed_at	TIMESTAMP NULL	Completion time.	2026-07-01T08:02:15Z

Recommended indexes:

- (fulfillment_id, status).
- (command_ref).

5.3 equipment_swap_fulfillment_reconciliation

What it is for: stores post-provisioning verification between BSS and network state.

When to use it: after provisioning commands finish, especially when external systems are eventually consistent or manually operated.

Sample meaning: esfr-001 confirms network now shows new ONT serial bound to the subscription/HomePass.

Column	Type	Purpose	Sample
reconciliation_id	TEXT PK	Reconciliation id.	esfr-001
fulfillment_id	TEXT	Parent fulfillment.	esf-001
check_code	TEXT	Check identifier.	TARGET_ONT_BOUND
expected_value_json	JSONB	Expected state.	{"serial":"NEW123"}
actual_value_json	JSONB NULL	Observed state.	{"serial":"NEW123"}
status	TEXT	PASS, WARN, FAIL.	PASS
message	TEXT	Human-readable result.	Target ONT bound in OLT.

Column	Type	Purpose	Sample
checked_at	TIMESTAMP	Check timestamp.	2026-07-01T08:05:00Z

Recommended indexes:

- (fulfillment_id, status).

5.4 equipment_swap_exception

What it is for: stores exception/manual repair cases for swap fulfillment.

When to use it: provisioning mismatch, missing network response, serial mismatch, forced completion, rollback decision, or manual override.

Sample meaning: ese-001 records that HFC authorization failed and a NOC user must repair it.

Column	Type	Purpose	Sample
exception_id	TEXT PK	Exception id.	ese-001
fulfillment_id	TEXT	Parent fulfillment.	esf-001
exception_type	TEXT	PROVISIONING_FAILED, SERIAL_MISMATCH, NETWORK_MISMATCH, FORCE_COMPLETE, ROLLBACK_REQUIRED.	PROVISIONING_FAILED
severity	TEXT	LOW, MEDIUM, HIGH, CRITICAL.	HIGH
status	TEXT	OPEN, PENDING_APPROVAL, APPROVED, RESOLVED, CANCELLED.	OPEN
approval_request_id	TEXT NULL	EM-CFG-04 request if needed.	apr-esf-001
assigned_to_team_code	TEXT NULL	Repair team.	NOC_GPON
created_at	TIMESTAMP	Creation timestamp.	2026-07-01T08:06:00Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	null

Recommended indexes:

- (fulfillment_id, status).

- (approval_request_id).

6. APIs

6.1 Start Swap Fulfillment

```
POST /api/equipment-swap-fulfillments
Idempotency-Key: esf-swap-001
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "swapRequestId": "swap-001",
  "woId": "wo-001",
  "subscriptionId": "SUB-700045",
  "accountId": "ACC-900045",
  "swapType": "DEFECT_SWAP",
  "technologyFamily": "GPON",
  "sourceInstanceId": "inst-old-001",
  "targetInstanceId": "inst-new-001",
  "homepassId": "HP-001"
}
```

6.2 Execute Fulfillment

```
POST /api/equipment-swap-fulfillments/{fulfillment_id}/execute
Content-Type: application/json
```

Execution:

- validate OSR-RMA swap and OSR-INSTANCE source/target state;
- call PROV-INT/FUL-03 adapters for old unbind/new bind;

- update HomePass path if needed;
- reconcile network state;
- notify OSR-RMA of fulfillment result.

6.3 Fulfillment Callback To OSR-RMA

```
POST /internal/swap-requests/{swap_request_id}/fulfillment-outcome
Content-Type: application/json

{
  "fulfillmentId": "esf-001",
  "outcome": "COMPLETED",
  "sourceInstanceId": "inst-old-001",
  "targetInstanceId": "inst-new-001",
  "completedAt": "2026-07-01T08:08:00Z"
}
```

6.4 Manual Exception Approval

For force-complete or risky override, FUL-09 calls EM-CFG-04:

```
POST /api/approvals/evaluate
Content-Type: application/json

{
  "operatorCode": "WIK",
  "moduleCode": "FUL-09",
  "actionCode": "EQUIPMENT_SWAP_FORCE_COMPLETE",
  "subjectType": "EQUIPMENT_SWAP_EXCEPTION",
  "subjectId": "ese-001",
  "context": {
    "fulfillmentId": "esf-001",
    "exceptionType": "NETWORK_MISMATCH",
    "technologyFamily": "GPON",
    "subscriptionId": "SUB-700045"
  }
}
```

7. Technology-Specific Behavior

Technology	Old equipment action	New equipment action	Reconciliation
GPON	Unbind old ONT from OLT/NMS; clear old service binding.	Bind/provision new ONT serial; update HomePass network path if changed.	OLT shows target serial/service active.
HFC	Disable old modem/STB/smart-card authorization; preserve slot/port where required.	Authorize new MAC/smart card; update Broadhub/NOMS/Verimatrix as applicable.	CMTS/Broadhub/entitlement state matches BSS.
IPTV	Remove old STB/smart-card entitlement.	Add entitlement to new STB/smart-card.	Entitlement system shows new device.
VOICE	Unbind old MTA/IAD port/number mapping.	Bind phone numbers to new port/device.	Voice platform shows new binding.

8. Business Rules

Rule	Description	Reason
R-FUL-09-01	Swap request must exist and be in a fulfillment-ready state.	Avoid orphan provisioning.
R-FUL-09-02	Target instance must be reserved/assigned by OSR flow before provisioning.	Inventory ownership.
R-FUL-09-03	FUL-09 must not call OSR-02 procurement during customer job.	Procurement is upstream replenishment only.
R-FUL-09-04	Old device unbind must happen before new device bind unless operator policy allows make-before-break.	Avoid duplicate network bindings.
R-FUL-09-05	Reconciliation must pass before completion callback, unless approved force-complete is used.	Service integrity.

Rule	Description	Reason
R-FUL-09-06	Manual force-complete/override must use EM-CFG-04 when policy requires it.	Governance.
R-FUL-09-07	Camunda approval process is owned by EM-CFG-04; FUL-09 owns its fulfillment workflow.	Clear process boundaries.
R-FUL-09-08	All provisioning commands must be idempotent by fulfillment/step id.	Safe retries.

9. Rule Engine Guidance

Use Drools for configurable swap fulfillment decisions:

Rule file	Purpose
equipment-swap-fulfillment-routing.drl	Select provisioning steps by technology, SKU, service family, and operator.
equipment-swap-make-before-break.drl	Decide whether old/new can overlap during swap.
equipment-swap-exception-approval.drl	Decide when manual override needs EM-CFG-04 approval.

Example:

```
package com.sophix.fulfillment.equipment_swap

rule "GPON target bind requires OLT reconciliation"
when
    $f : EquipmentSwapFulfillmentFact(technologyFamily == "GPON")
    $d : EquipmentSwapFulfillmentDecision()
then
    $d.addStep("GPON_UNBIND_OLD_ONT");
    $d.addStep("GPON_BIND_NEW_ONT");
    $d.addReconciliationCheck("TARGET_ONT_BOUND");
end
```

10. Events

Event	Topic	When emitted
EquipmentSwapFulfillmentRequested	sophix.fulfillment.equipment-swap.requested	Fulfillment created.
EquipmentSwapProvisioningStepCompleted	sophix.fulfillment.equipment-swap.step-completed	A provisioning step completes.
EquipmentSwapFulfillmentCompleted	sophix.fulfillment.equipment-swap.completed	Fulfillment complete.
EquipmentSwapFulfillmentFailed	sophix.fulfillment.equipment-swap.failed	Fulfillment failed.
EquipmentSwapExceptionOpened	sophix.fulfillment.equipment-swap.exception-opened	Manual repair/approval needed.

11. Frontend/API Consumers

Area	APIs/modules used
Backoffice NOC/support	FUL-09 fulfillment detail and exception APIs.
Contractor mobile app	WO-01 task execution; not direct provisioning except finalization data.
OSR-RMA module	Starts FUL-09 and receives outcome callback.
Relocation/shifting flows	Consume FUL-09 when new/recovered equipment requires provisioning changes.
Reporting	FUL-09 events into REP-01.

12. Infrastructure Recommendations

Run FUL-09 inside fulfillment/provisioning API and worker deployments.

Component	Recommendation
API	Fulfillment API exposes swap-fulfillment endpoints.
Workers	Shared fulfillment/provisioning worker executes provisioning steps.
Database	Fulfillment schema.
Provisioning	Use PROV-INT-01 adapters and reconciliation.
Events	Kafka/outbox for FUL-09 events and OSR-RMA callbacks.
Rules	Drools/KIE for configurable technology behavior.

AWS notes: use existing ECS/EKS fulfillment services, RDS PostgreSQL, MSK/Kafka, Secrets Manager for provisioning credentials, and strict alerts for failed provisioning/reconciliation.

13. Acceptance Criteria

- Swap fulfillment tables and APIs exist.
- OSR-RMA can start fulfillment and receive outcome.
- FUL-09 validates OSR-RMA and OSR-INSTANCE state through APIs.
- Provisioning commands are executed through PROV-INT/FUL-03 adapters.
- GPON/HFC/IPTV/VOICE technology rules are configurable.
- Manual force-complete uses EM-CFG-04 when configured.
- Procurement is explicitly not in the customer swap runtime path.
- Tests cover success, provisioning failure, reconciliation failure, forced completion approval, and idempotent retry.